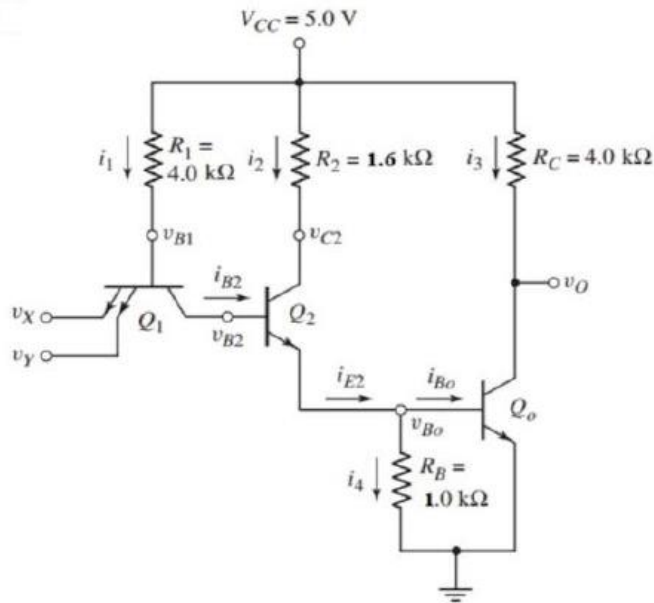


Exercise 1



For the TTL NAND circuit, assume $\beta_F = 25$, $\beta_R = 0.1$

$V_{OH} = 3.4\text{ V}$, $V_{OL} = 0.1\text{ V}$

For saturation mode, assume, $V_{CE} = 0.1\text{ V}$

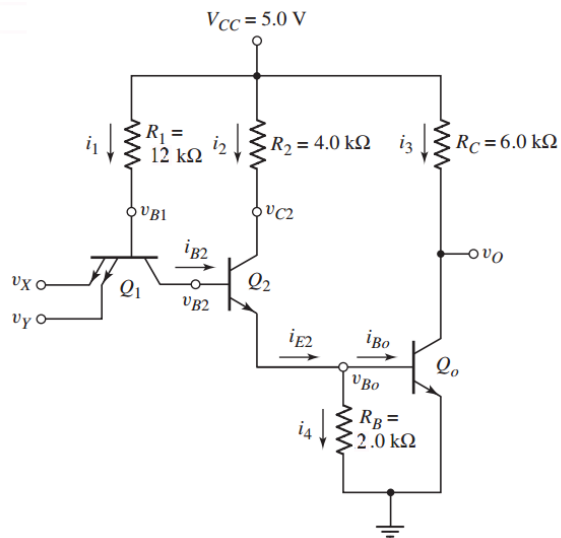
- Assume no loads are connected to the driver device. If at least one input is low (0.1 V), find i_1 , i_{B2} , i_{Bo} , i_3 .
- Repeat the calculation of (a) if both inputs are high ($v_x = v_y = 5\text{ V}$).
- Find the maximum possible fanout if the loads are also this TTL NAND gate.
- Assume all inputs of the **driver** are **high**.
 - Find the maximum fanout if the other input of the **load** $v_y = 5\text{ V}$ (**High**).
 - If **both inputs** of the **load circuits** are **low**, then what would be the maximum fanout?

Comment and compare on the above two cases to identify which case has better fanout and why?

- Find the power dissipation for all cases [Assume 4 loads are connected to the driver's output]

Ans: a) 1.025 mA , all other currents 0 mA
 b) 0.675 mA , 0.81 mA , 2.5725 mA , 1.225 mA
 c) 5
 d) 1.61 , 2.123
 e) 23.397 mW

Practice Problem 1



For the given TTL circuit, $\beta_F = 25$, $\beta_R = 0.1$. Assume any of the input to the driver is low and loads are not connected.

- Find i_1 .
- Find i_{B2} .
- Find i_2 .
- Find i_{B0} .
- Find i_3 .

Ans: a) 0.342 mA

b) 0 mA

c) 0 mA

d) 0 mA

e) 0 mA

Now consider all input voltages high, find the same current values again.

Ans: a) 0.225
b) 0.27
c) 1.025
d) 0.895
e) 0.8167